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Welcome from Firstname Lastname, Conference Chair

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Conference Program

Proc. of the Xth International Students Workshop (ISW), Kamienna Góra, Poland, [Dates]

Multi-label music genre recognition based on spectrograms

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Abstract. In the current day and age, music surrounds us everywhere and accurate genre classification is important to both the recipients as well as music labels and stores. However, due to many factors, nowadays the genres of songs and albums are not clearly defined, often being mixed on a single record, which would require an approach different from the standard multi-class on, which many of the currently available datasets lack. In addition, they often suffer from the lack of proper annotations, insufficient size, and inadequate representation of the music landscape. That is why in this work an extensive comparison of various preprocessing methods and ways of handling imbalanced data is performed for a multi-label classification. For the experiments different types of spectrograms extracted from audio files were used, and the dataset was personally acquired and labeled based on the biggest on-line music database, resulting in more than 18000, multi-label and representative of current trends data.

Keywords: Multi-Label Classification · Acoustic Data Classification · Imbalanced Data Classification

1 Introduction

1.1 Multi-label nature of music

Music has accompanied humans throughout history, starting with primitive sounds to beautifully complex, classical compositions. However, in modern day and age, the number of new releases each year is constantly increasing. Looking, for example, at one of the biggest online music databases, namely Discogs, we can see how in the 1950s it was around 400 thousands and in the 2010s: more than 4 million. With increased numbers also came increased diversity in the sense of genres. New technologies, especially computer software, drastically changed what can be achieved and created, leading to more and more experimentation, and as a result: subgenres. Oftentimes they start as a slight variation, in time gaining more noticeable, distinct features that differentiate it from its predecessors. Sometimes, it is a result of cross-genre mixing, for example, a rock song with incorporated jazz instrumentation. In such instances, if we were to try to match that release to one more general genre, a case can be made for both. Such cases are also not obscure: A great example of this in recent years was Old Town Road by Lil Nas X and Billy Ray Cyrus, which features a rare combination of Hip-Hop and Country music. Some others, while not as mainstream-popular, bring forth innovation and create whole trends through unorthodox genre mixing. One such example is a Sacramento-based group Death Grips, who have shaped the entire 2010's in the hip-hop underground, through their mix of Hip-Hop, Industrial, Punk, and Experimental. Their influence has even transcended that underground, as it is unquestionable that their works have been an influence for Kanye West's 2013 album, Yeezus, which has proven to be a commercial and mainstream success. That is why, in the current landscape of music, limiting the belongingness to only one genre is a far too great oversimplification and thus is counterproductive. However, that leads to a completely new issue: data.

1.2 Problems with music datasets

For years in the field of Music Information Retrieval (MIR), the datasets used for various tasks come with different sets of problems. Some have a really small number of samples and a multitude of other issues[2](GTZAN[3]), some lack raw music files (Million Songs Dataset[4]), and others lack proper annotations (MagnaTagATune Dataset). Two other, major issues present in most, if not all, datasets is that they are inadequate for multi-label classification due to their tagging policy, and the tags present are of questionable quality and source (like Amazon shop). Another important aspect is the proper representation: in order to provide most value to the real-life scenarios, the data should reflect the existing landscape: containing modern music with adequate representation of popular genres. Thus, the basis for these experiments comes from the self-aggregated dataset which addresses each of the aforementioned problems. It contains more than 1400 albums, amounting to over 18000 songs, which span all the way back from the 1950s up to 2024, with the heaviest emphasis put on the currently dominant genres, like hip-hop, electronic, rock, and pop, while also having a representation of more niche genres, like industrial, folk, or ambient. In total, 16 major genres were distinguished, and each song was paired with up to three different ones. However, as with the real distribution, this dataset can be characterized as imbalanced and thus a difficult one. The most popular and numerous being hip-hop, consisting of around 5600 songs, while jazz is only 237. All the annotations have been obtained from one of the biggest online music databases: RateYourMusic. Thanks to its vast user base, which helps with tagging all the data, it is probably one of, if not the best source of detailed information about the releases. As the data collected are in the form of raw music files, consisting of mostly MP3, but also MP4 and FLAC files, any kind of information can be retrieved and any transformations can be applied, as needed.

2 Related works

The task of classifying music genres has been approached in many ways over the years. Tzanetakis and Cook [3] was one of the first works to show the music genre

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classification as a pattern recognition task. The authors proposed three sets of characteristics: timbral texture, rhythmic content, and pitch content. The classification itself was performed using Statistical Pattern Recognition, Gaussian, Gaussian Mixture Models, and k-Nearest Neighbors. Ndou et al. [5] conducted a case study of the performance of different classification approaches on the popular GTZAN dataset. Traditional machine learning techniques such as linear logistic regression, support vector machines, random forest, naive Bayes, or k-Nearest Neighbors were compared against each other and against a Convolutional Neural Network. The CNN was also trained in 3 different ways: by using 3 second fragments, 30-second fragments, and spectrograms. The CNN models were unfortunately underwhelming, which, as noted by the authors, may have been influenced by the computational constraints for the research. In Lo and Lin [6] showcases the usage of features more known in music theory, such as chords, melody, pitch, or even the frequency of occurrence of specific notes to perform the classification tasks on classical music.

The use of spectrograms was discussed by Nirmal and Mohan [7], who compared a self-made newly trained sequential CNN model with a pre-trained model VGG16. In Bahuleyan [8] the usage of CNN models in spectrograms, machine learning models in time domain features, and the combination of both was presented. Here also the VGG16 was used, and for the Machine Learning Algorithms the Linear Regression, Gradient Boosting, Random Forest, and Support Vector Machines. For the ensemble, the combination of VGG16 and Gradient Boosting was used and it achieved the best results of all. Costa et al. [9] uses and tests representation learning together with Convolutional Networks. In addition to using the spectrograms to perform the classification task, the authors also experimented with the fusion of both learned and handcrafted features to assess the complementarity of those two approaches.

The use of a CNN and RNN hybrid as Convolutional Recurrent Neural Networks in Choi et al. [10] shows some strong performances. CRNN is a modified CNN model, where the last convolutional layers are replaced with the Recurrent Neural Network model. Silla et al. [11] proposes the combination of using the time and space decomposition with ensemble classifiers. K-Nearest Neighbors, Naïve-Bayes, Decision Trees and Support Vector Machines were used on the novel Latin dataset. Schindler and Knees [16] tackle the multi-label problem of tagging music by features such as genre, style, mood or theme. Their Deep Neural Models were based on the triplet-network architecture. One of the earlier works on the multi-label music genre classification is Lukashevich et al. [12], which describes different approaches of multi-labeling: for the whole song, by dividing it into separate segments and by dividing into segments while simultaneously dividing into domains, such as Timbre, Rhythm, Melody, and Harmony. A fairly similar approach to detect genre borders was proposed in Nakamura et al. [14], which further adds the possibility of applying a single label classification on each detected segment in order to achieve a multi-label classification for the entire song.

In Oramas et al. [13], the authors present a comparison of different feature extraction methodologies from audio, text reviews, and image covers, as well as their combinations, showing ways to improve accuracy by implementing approaches that may not sound so obvious at first. Here, the CNN is used on the spectrograms for audio, a Vector Space Model for the text, and a modified version of the ResNet model for the covers. The results show the potential of the multimodal approach. Sanden and Zhang [15] discuss different ensemble techniques for the specific task of the multi-label classification of music genres. Although using only a simpler machine learning approach, the whole research is very robust, covering multiple datasets and ensemble techniques.

3 Experimental evaluation

This section describes the evaluation process designed to answer the following questions:

– How different methods of imbalance handling impact the multi-label classification?
– How do different methods of preprocessing the audio data impact the multi-label classification?

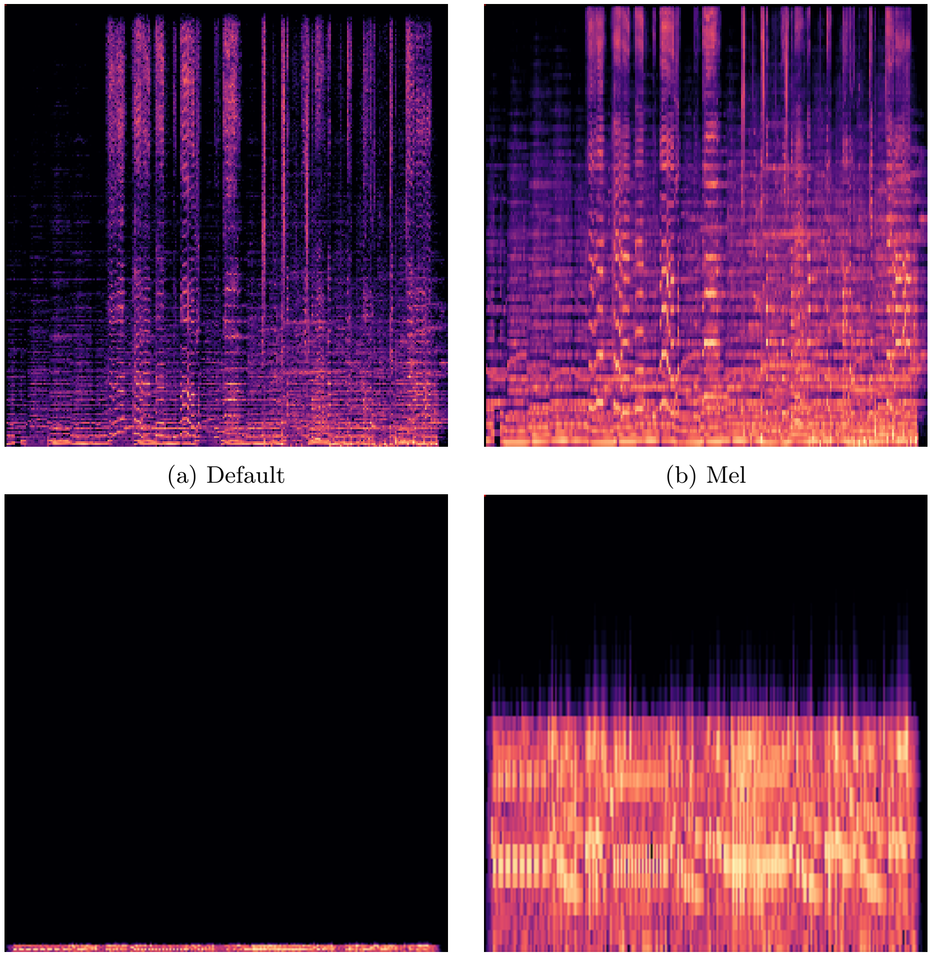
3.1 Set-up

Dataset The basis for the experiments was a self-assembled dataset of songs in different formats (MP3, MP4 or FLAC). Each of the 18019 songs has between 1 to 6 genre tags attached, which were taken from largest community-maintained online music database. The table 1 shows the distribution of songs between genres.

Table 1: Number of songs by genre

Hip-Hop	5641	Electronic	4680	Rock	4356	Pop	2758
Psychedelia	1367	Metal	1153	Dance	960	Punk	804
R&B	801	Industrial & Noise	788	Experimental	694	Ambient	687
Folk	652	Classical Music	608	Singer-Songwriter	474	Jazz	278

Data preprocessing The data in the experiment is analyzed in two forms: spectrograms (Figure 1a) and mel scale spectrograms (mel spectrograms) (Figure 1b). Spectrogram is a visual representation of signal frequencies over time, and mel scale is a perceptual scale adjusted to human hearing [1]. Additionally, a spectrogram obtained from a reduced audio file (Figure 1c), and the same spectrograms scaled to fill more space of the same reduced file (Figure 1d) were used.



(a) Default (b) Mel

(c) Reduced (d) Reduced and scaled

Fig. 1: Sample spectrograms of different preprocessing approaches

Model architecture The architecture was an ensemble of 16 binary classification models, one per genre, each trained using a *One-vs-All (OVA)* binarization strategy. The model underneath was a custom, self-implemented CNN model.

3.2 Experimental scenario

Scenario Comparison of different approaches to handling the imbalanced nature of the data: no imbalance handling, undersampling with shuffling throughout the learning epochs, and shuffling-free stratified undersampling.

Goals Analysis of impact of different techniques to handling the imbalanced dataset problem on the multi-label dataset

Scenario Comparison of different approaches to preprocessing the audio files before extracting spectrograms: default files, a reduced audio file, and reduced file scaled. The reduction was performed by taking only one every 64 frequency values at each point in time, and taking the median of it. The effect is an echo-y audio, resembling the original if one knows what it should sound like, otherwise being rather undistinguishable. The scaling applied in the last approach was to simply scale on the y-axis to simply fill more of the space.

Goals Analysis of impact of an artificial, lossy transformation of the audio data on the multi-label music dataset

3.3 Evaluation Protocol

Due to the size of the data set and the number of models that had to be trained, it was ultimately decided not to perform any cross-validation. On the basis of one run, several metrics were used. For singular-genre models: Accuracy, F1 Score, Balanced Accuracy, Precision, and Recall.

3.4 Reproducibility

The code was written using *Python* 3.12.7, with CUDA version 12.4. The libraries used included *TensorFlow* 2.16.1, *Librosa* 0.10.2, *scikit-learn* 1.4.2, *Pandas* 2.2.2 and *Matplotlib* 3.9.0, using a computer with 64GB of RAM, a 16-core CPU, and 8448 CUDA cores GPU with 16GB of VRAM available.

4 Results

In the following diagrams, different approaches are described as following:

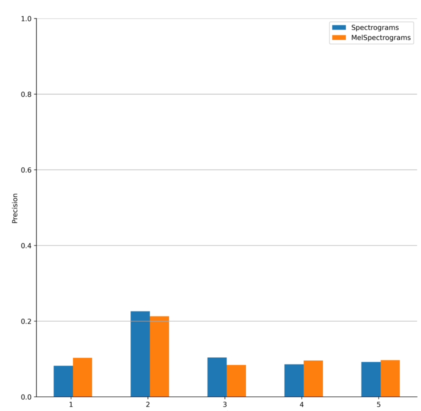


Fig. 2: Precision scores of different approaches

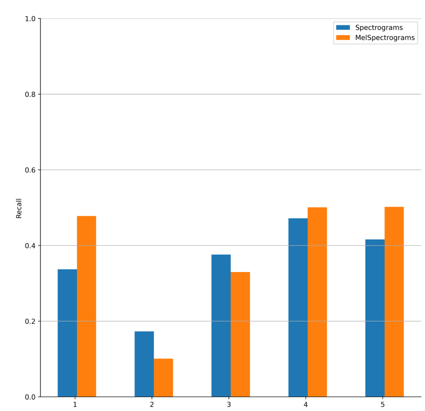


Fig. 3: Recall scores of different approaches

shuffling through epochs - experiment 1; no imbalance handling - experiment 2; stratified undersampling - experiment 3; reduced audio with stratified undersampling - experiment 4; reduced and scaled audio with stratified undersampling - experiment 5.

On figures 2 and 3, with precision and recall respectively, we can see that all except of the second approach perform similarly, especially with regards to precision. In 3 out of 5 also the mel spectrograms perform slightly better. In the approach with no handling of the imbalance, while precision is noticeably higher than the others, it's a tradeoff with the recall.

Thus when taking a look at figure 4 with F1 Score, it all evens out across the board, and also at figure 5 with the Balanced Accuracy Score, the previous trends of mel spectrograms outperforming the default ones in approaches 1, 4 and 5 continue, as well as the seemingly best scores of approach 2. This is most probably a result of the number of total examples provided to the models at the training time, as due to hands-off approach in the second experiment, all of available training examples were provided, whereas in other approaches only some (rather small) part was used. This proves that another approach, rather than simple stratified undersampling should be used to achieve better results while maintaining the focus on the minority class in each case.

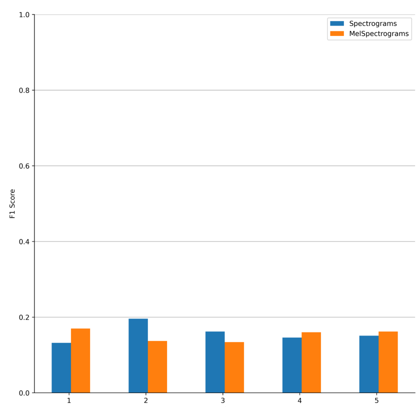


Fig. 4: F1 scores of different approaches

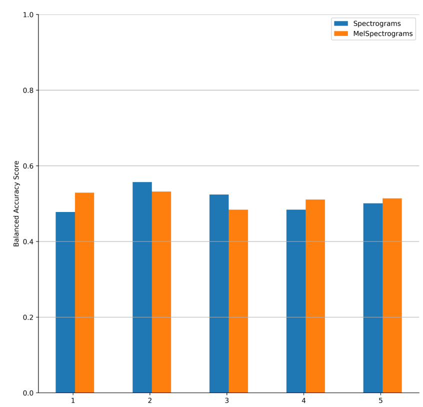


Fig. 5: Balanced Accuracy scores of different approaches

However, I believe that most interesting is the comparison between the normal spectrograms and ones achieved from reduced audio files with scaling applied. Looking at the former 6 and the latter 7 figures, the scores achieved across the board are very similar, with the biggest differences being visible in specificity, where the normal files prevail, and in recall, where the situation is turned. Interestingly, the resulting aggregate score such as F1 Score and Balanced Accuracy are very similar, and shows, that even with a lossy transformation of the audio data, the machine learning models are able to extract the features necessary for successful classification no worse than from normal, much larger audio files.

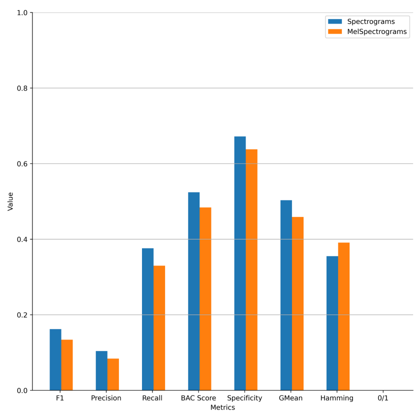


Fig. 6: Different scores in the third approach

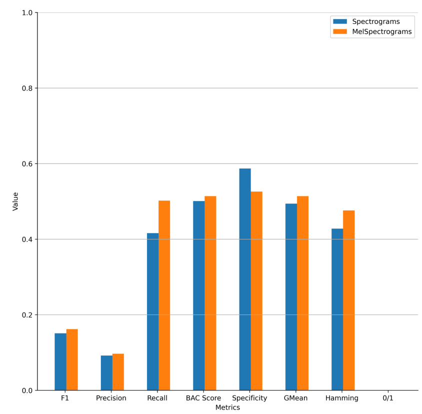


Fig. 7: Different scores in the fifth approach

5 Conclusions and future research

The general scores achieved by the trained models are rather underwhelming - the aggregate scores such as F1 being in the range between 0.1 and 0.2. This can be most likely attributed to the simplistic nature of the CNN architecture used for the experiments, and thus easily fixable in the future works, by utilizing pretrained and more complex state-of-the-art models such as Resnet or VGG. Despite that fact the conclusions that can be extrapolated from this work are very interesting: the artificial and lossy preprocessing of the audio files can bring competitive results when compared with standard audio files. However, the question of comparison of default and mel spectrograms still stands: more robust models would probably bring more definitive results in that matter. An interesting direction where the future research could be taken would be to perform similar experiment on different state-of-the-art non-CNN solutions, like transformers and state space models, namely Audio Spectrogram Transformer [18] and Audio Mamba [17].

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Eye opening detection in early neonatal period through CNN

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Abstract. Automated detection of neonatal eye opening could support continuous neurological monitoring and facilitate timely clinical interventions in Neonatal Intensive Care Units (NICU). However, this task is challenging owing to unique NICU conditions such as low-contrast lighting, medical device obstructions, and neonates' diminutive facial anatomy. This study explored deep learning approaches for automated neonatal eye state detection by comparing a lightweight CNN architecture trained from scratch with four transfer learning models: VGG16, DenseNet121, ResNet101, and EfficientNetB0. The evaluation was conducted on a carefully curated dataset of 4,119 labeled facial frames extracted from 46 clinical NICU videos. EfficientNetB0 achieved the highest performance with an AU-ROC of 0.928, demonstrating its excellent capability to distinguish between open and closed eyes. The lightweight CNN also yielded competitive results with a significantly shorter training time, highlighting its potential for real-time clinical monitoring. Future research directions include dataset expansion, incorporation of additional facial cues, and application of explainability methods to further enhance clinical interpretability and robustness.

Keywords: Neonatal Monitoring · Eye Opening · Deep Learning · Convolutional Neural Networks · Transfer Learning

1 Introduction

Monitoring whether a neonate's eyes are open or closed is clinically relevant for several reasons. First, eye opening is a direct behavioral marker of arousal and responsiveness, routinely scored in neurological examinations such as the *Neonatal Behavioral Assessment Scale* and *Sarnat* staging of hypoxic-ischemic encephalopathy [1]. Second, fluctuations in the eye state correlate with nociceptive responses and the depth of pharmacological sedation, helping clinicians regulate analgesics and avoid pain under- and over-sedation [2]. Finally, because most life-support equipment in the Neonatal Intensive Care Units (NICU) obscures other facial cues, the eyes remain a comparatively accessible window into the neonate's neurological integrity [3].

Despite its clinical importance, continuous manual observation is impractical in busy NICUs and is subject to considerable inter-observer variability. Vision-based automation offers a promising alternative, enabling objective high-frequency measurements while minimizing handling, a factor known to reduce stress, conserve energy reserves, and promote neurodevelopmental stability.

In recent years, this automation has been increasingly powered by advances in computer vision and Deep Learning (DL), particularly the emergence of pre-trained Convolutional Neural Networks (CNN), which have demonstrated strong performance in face detection and landmark localization tasks in adults [5]. When integrated into multimodal monitoring systems alongside signals, such as electroencephalography or vital signs, real-time eye state information can enrich early warning systems and support timely neuroprotective interventions [6].

However, neonates present unique challenges: diminutive facial anatomy, variable head posture, medical devices (nasal cannulae, phototherapy goggles), and low-contrast lighting common to incubators. Although domain-specific detectors, such as *NICUface*, have begun to address these constraints, frame-level eye state classification in neonates remains largely unexplored, with existing approaches relying on small datasets and handcrafted heuristics [4].

In this study, an end-to-end framework is presented that first crops the infant's face with an NICU-adapted detector and then classifies the eye state using two alternative CNN-based strategies: (i) a lightweight model trained from scratch and (ii) a transfer learning baseline. The approach was evaluated using a curated dataset of 4,119 manually annotated frames extracted from 46 clinical videos recorded during routine neurological assessments.

The remainder of this paper is organized as follows: Section 2 describes the dataset and the applied preprocessing steps; Section 3 presents the proposed classifiers; Section 4 details the training and evaluation protocol; Section 5 reports and discusses the results; and Section 6 summarizes the main findings and outlines future work.

2 Dataset

A total of 46 NICU videos were collected from 17 neonates during their first days of life at the *Hospital Universitario de Burgos*. All participants experienced a hypoxic event during birth; the group included both preterm (born between 34 and 36 weeks of gestation) and term infants. Informed consent was obtained from the legal guardians of all participants prior to participation. The recordings underwent the following preprocessing pipeline.

- **Video standardization and clip segmentation.** The original files (*mov*, *avi*, and *mp4*) were converted to *mp4* and split into 400 short clips that isolated distinct clinical stimulations.
- **Frame extraction.** Frames were extracted from each clip at 30 frames per second (fps).

- **Face detection**. Faces were located with the MTCNN detector from foundation model called *FaceNet-PyTorch*[3], retaining only the highest-confidence bounding box per frame. Frames without valid detection were discarded.
- **Temporal down-sampling.** For every 1 second window, the frame with the highest detection confidence was selected, ensuring sharp and centered facial views and reducing redundancy from nearly identical consecutive frames.
- **Face cropping and normalization**. The selected bounding box was expanded to a square, resized to 224×224 pixels, and its pixel values were normalized to the mean and standard deviation used for the pretrained models.
- **Manual labeling**. Each retained frame was manually categorized as *open* or *closed*. Blurred, occluded, or poorly lit samples were discarded. The resulting dataset contains 4,119 labeled face crops, 3,194 open (≈77%) and 925 closed (≈23%), yielding a class ratio of approximately 3.5:1 that is explicitly addressed in the training phase. The distribution is shown in Figure 1.

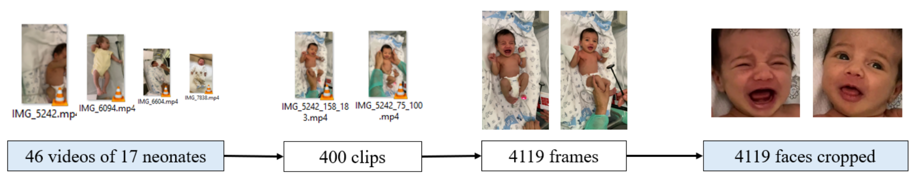

Fig. 1. Overview of the preprocessing pipeline yielding the 4,119 labeled face crops used for model training and evaluation.

3 Methodology

A supervised learning approach is employed by leveraging the labeled dataset described in Section 2. Two complementary CNN-based classification strategies were explored: (i) training a lightweight CNN from scratch, inspired by the classical AlexNet architecture, and (ii) applying transfer learning using several pre-trained state-of-the-art architectures.

3.1 CNN trained from scratch

In the first strategy, a lightweight CNN was specifically designed to balance the computational efficiency and classification accuracy. The architecture was loosely inspired by AlexNet, a pioneering deep CNN composed of five convolutional layers followed by three fully connected layers, demonstrating the potential of DL

[3] `https://github.com/timesler/facenet-pytorch`

for image recognition tasks [7]. Similar to AlexNet, the proposed CNN includes multiple convolutional layers interspersed with max-pooling operations, employs ReLU activation functions, and culminates in fully connected layers for the final classification. However, modifications, such as fewer filters per convolutional layer and reduced fully connected layer dimensions, were introduced to limit the complexity and computational demands, reflecting the smaller scale of the neonatal dataset used in this study.

3.2 Transfer learning

Transfer learning leverages the knowledge gained from large-scale datasets (ImageNet in this case) to facilitate training on smaller specialized datasets, thereby reducing the risk of overfitting and significantly accelerating model convergence [8]. In this study, four popular pre-trained CNN architectures were evaluated.

- **VGG16**. Known for its simplicity and effectiveness, this architecture consists of 16 convolutional layers using small (3×3) kernels stacked sequentially, followed by max-pooling operations [9]. Although it is computationally demanding owing to its dense layers, it remains a popular baseline for transfer learning due to its robust learned representations.
- **DenseNet121**. This is notable for its dense connectivity pattern, where each layer receives input from all preceding layers and passes its outputs to all subsequent layers, thereby significantly alleviating the vanishing gradient problem [10]. This architecture efficiently reuses features, requires fewer parameters, and offers competitive performance for various vision tasks.
- **ResNet101**. Utilizes residual connections (skip connections) to address degradation problems in deep networks, enabling the stable training of very deep models with 101 layers [11]. By explicitly modeling residual mappings, ResNet facilitates deeper network architectures without compromising accuracy, thereby leading to improved representational capacity.
- **EfficientNetB0**. Introduces a novel compound scaling method that systematically balances depth, width, and input image resolution to enhance accuracy while maintaining low computational requirements [12]. EfficientNetB0 is the baseline model of the EfficientNet family and demonstrates high performance in image classification tasks with significantly fewer parameters than traditional CNNs.

For each of these pre-trained models, fine-tuning was conducted by initially freezing most of the early convolutional layers (retaining learned low-level features) and retraining only the deeper convolutional and fully connected layers. Importantly, the number of frozen layers was consistent across all transfer learning architectures. This strategy preserved previously acquired generalizable knowledge, adapting only the most specific task-relevant representations for neonatal eye state classification.

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4 Experimental Design

4.1 Training strategy

The experimental evaluation followed a subject-independent 5-fold cross validation scheme. In each iteration, one fold was reserved for testing, another for validation, and the remaining three were used for training. To prevent data leakage and ensure unbiased evaluation, all frames from the same neonate were consistently assigned to a single fold. In addition, the folds were balanced to maintain approximately similar proportions of *open* and *closed* labels.

All CNN models were trained using the Adam optimizer with a batch size of 64 and an input image size of 224×224 pixels, which is consistent with the standard dimensions established for the pretrained architectures. Early stopping was implemented to monitor the area under the ROC curve (AU-ROC) of the validation set to halt training once the performance plateaued. Binary cross-entropy was selected as the classification loss function. To address the class imbalance, class weights were computed and applied during training, assigning greater importance to the underrepresented (*closed*) class.

Training was conducted using an InfraRed GPU cluster provided by *Junta de Castilla y León*, equipped with 30 NVIDIA Tesla A100 GPUs (40 GB memory each) and delivering a peak computing capacity of 9360 TFLOPS (FP16).

4.2 Evaluation metrics

Model performance was assessed using standard classification metrics: accuracy, F1-score, and geometric mean (G-mean). In addition, the areas under the precision-recall (AU-PR) and receiver operating characteristic (AU-ROC) curves were calculated. Accuracy provides overall predictive correctness; F1-score balances precision and recall, reflecting effectiveness in imbalanced settings; and G-mean evaluates balanced predictive performance between classes. AU-PR and AU-ROC further quantify each model's discriminatory ability independent of classification thresholds, which is especially useful in imbalanced scenarios such as this one.

5 Results and discussion

Table 1 summarizes the average performance metrics computed over the 5 folds for each tested model, presenting the results from the best-performing hyperparameter configurations. The lightweight CNN trained from scratch demonstrated strong performance, with an AU-ROC of 0.908 and AU-PR of 0.974, achieving a notable accuracy of 0.904 and F1 score of 0.945. Notably, this configuration required only 1.78 minutes of training time, underscoring its computational efficiency and potential suitability for near real-time clinical monitoring.

Among the transfer learning strategies, EfficientNetB0 yielded the best performance, achieving the highest AU-ROC (0.928) and a notably high AU-PR

Table 1. Performance metrics (Accuracy, F1 score, G-mean, AU-PR, and AU-ROC) averaged over 5 folds for the CNN trained from scratch and four transfer learning architectures. Training times (in minutes) per model are also reported.

Metrics	**CNN**	**DenseNet121**	**EfficientNetB0**	**ResNet101**	**VGG16**
Accuracy	0.904	0.869	**0.908**	0.876	0.891
F1 score	0.945	0.913	0.939	0.920	0.930
G-mean	0.823	0.789	0.833	0.759	0.780
AU-PR	0.974	0.969	**0.976**	0.966	0.981
AU-ROC	0.908	0.905	**0.928**	0.903	0.913
Time (min)	**1.782**	14.301	15.103	13.494	14.765

(0.976). Although the lightweight CNN achieved a marginally higher F1-score (0.945 vs. 0.939), EfficientNetB0 was prioritized because of its superior AU-ROC, the primary metric for imbalanced classification tasks, as it evaluates model performance across all decision thresholds. With an accuracy of 0.908 and a G-mean of 0.833, this model demonstrated a robust ability to effectively classify neonatal eye states, although at the expense of longer training times (15.1 min) than the CNN trained from scratch.

The remaining transfer learning models (ResNet101, DenseNet121, and VGG16) also exhibited competitive performance, with AU-ROC values consistently above 0.90, G-mean scores of 0.759–0.789, and accuracy of approximately 0.87–0.89, indicating their effectiveness despite slightly lower metrics. Overall, these results illustrate that both lightweight CNNs trained from scratch and pre-trained architectures offer viable approaches for reliable and efficient neonatal eye state detection, with the choice depending on clinical priorities.

Fig. 2. Comparison of the ROC (left) and PR (right) curves for the best performing fold of the chosen model.

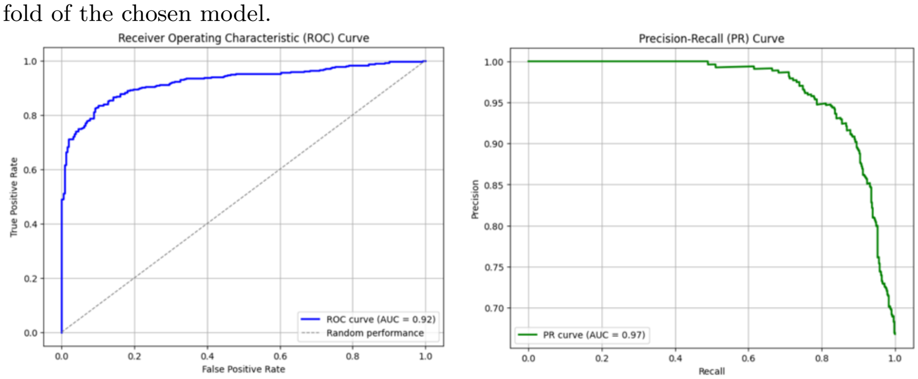

Figure 2 illustrates the ROC and PR curves for the best-performing fold of EfficientNetB0. The ROC curve reached an AU-ROC of 0.928, indicating excellent discriminative capability across a range of classification thresholds. In addition, the PR curve, with an AU-PR of 0.976, further emphasizes EfficientNetB0's strong performance under conditions of class imbalance, confirming its robustness as a suitable classifier for this application.

6 Conclusion and further work

This study introduced an automated CNN-based pipeline for frame-level classification of neonatal eye state, leveraging face detection and preprocessing specifically tailored to NICU video footage. Among the evaluated models, EfficientNetB0 demonstrated the strongest overall performance, providing an optimal balance between classification accuracy and computational efficiency. The AlexNet inspired CNN trained from scratch also achieved highly competitive results with a significantly reduced training time, highlighting its suitability for real-time deployment in clinical environments.

However, this work has several limitations. First, the class imbalance of the dataset (77% open-eye frames) may bias the model toward the majority class, despite the use of class weights. Second, all neonates in this study experienced hypoxic events, limiting the generalizability of the results to healthy populations. Finally, the pipeline's reliance on face detection introduces a potential failure point in cases of extreme head rotation or medical device occlusion.

Future research should focus on expanding the current dataset by incorporating videos from diverse clinical settings and patient populations, thereby enhancing the robustness and generalizability of the model. Extending the existing pipeline to incorporate additional facial cues, such as mouth opening, crying, or facial grimacing, could enable comprehensive multimodal neonatal monitoring and provide a more reliable assessment of neonatal attention and responsiveness. Exploring model explainability techniques, such as Grad-CAM or saliency maps, will also be pursued to identify the regions and visual patterns that influence model predictions, thus improving clinical interpretability and trust in the model. Finally, further hyperparameter optimization and ensemble strategies should be investigated to enhance the prediction consistency and real-time applicability of the system in clinical practice.

Ultimately, the developed CNN-based eye state detection pipeline represents a promising step towards automated, objective, and non-invasive neurological monitoring in NICU environments, potentially improving neonatal clinical outcomes through earlier interventions and more precise care.

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